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1.Socketserver

```
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GNU nano 8.7.1 socketserver.c
#include<stdio.h>
#include<unistd.h>
#include<sys/socket.h>
#include<sys/un.h>
#include<string.h>

#define SOCKET_PATH "/tmp/my_socket"
int main()
{
    int server_fd,client_fd;
    struct sockaddr_un address;
    char buffer[100];
    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);
    unlink(SOCKET_PATH);
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));
    listen(server_fd,5);
    printf("Server Waiting\n");
    client_fd = accept(server_fd,NULL,NULL);
    read(client_fd,buffer,sizeof(buffer));
    printf("Message from client: %s\n",buffer);
    close(client_fd);
    close(server_fd);
    unlink(SOCKET_PATH);
    return 0;
}
```

2.Socketclient

```
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GNU nano 8.7.1 socketclient.c
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>
#define SOCKET_PATH "/tmp/my_socket"
int main(){
    int client_fd;
    struct sockaddr_un address;

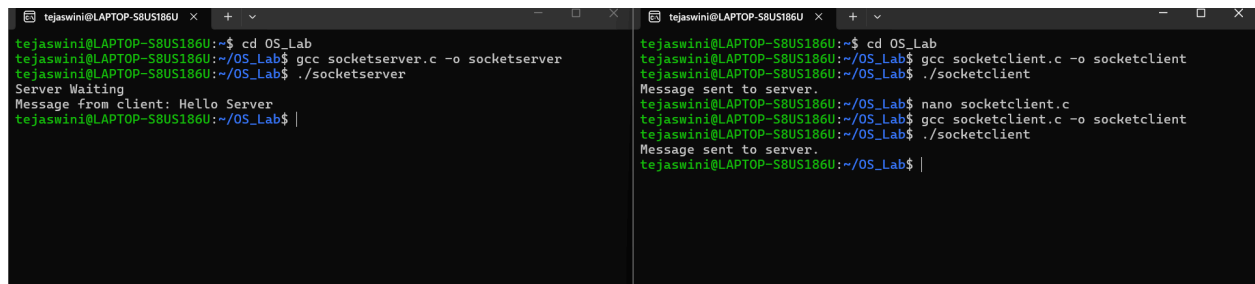
    client_fd = socket(AF_UNIX,SOCKET_STREAM,0);

    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    connect(client_fd,
            (struct sockaddr *)&address,
            sizeof(address));

    write(client_fd, "Hello Server",12);
    printf("Message sent to server.\n");
    close(client_fd);
    return 0;
}
```

Output:



The image shows two terminal windows side-by-side. The left window shows the compilation of a socket server and its execution, receiving a message from a client. The right window shows the compilation of a socket client, its execution, and the messages sent to the server.

```
tejaswini@LAPTOP-S8US186U x + v
tejaswini@LAPTOP-S8US186U:~$ cd OS_Lab
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc socketserver.c -o socketserver
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./socketserver
Server Waiting
Message from client: Hello Server
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |

tejaswini@LAPTOP-S8US186U x + v
tejaswini@LAPTOP-S8US186U:~$ cd OS_Lab
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc socketclient.c -o socketclient
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./socketclient
Message sent to server.
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano socketclient.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc socketclient.c -o socketclient
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./socketclient
Message sent to server.
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |
```